LESSON PLAN

COMPOSITE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Comparators, introduction types and their application
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Comparators, introduction types and their application
5. Specific Objectives	19.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Comparator operation, Safety: Handle delicate optical components with care, Instructor supervision required during setup
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a comparator in engineering metrology?
(b) Name two common types of comparators used in workshops.
(c) Why are comparators preferred over standard measuring instruments for high-precision work?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and working principle of comparators • Classification of comparators based on mechanism • Optical and mechanical comparator types • Key components of a standard comparator • Basic calibration procedures	L & Ds	PPT, IB	15 min
(b)	• Applications of optical comparators in inspection • Usage of dial indicators for dimensional checks • Selection criteria for specific comparator types • Error sources and mitigation techniques • Maintenance and storage requirements	L & Ds	PPT, PDF	20 min
(c)	• Demonstration of comparator setup and operation • Comparison of measurement accuracy across types • Troubleshooting common operational issues • Recording measurement data correctly • Safety protocols during practical handling	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between an optical comparator and a dial indicator in terms of application.
(b) Describe one method to minimize measurement errors when using a comparator.
(c) How would you select the appropriate comparator type for inspecting a complex profile?

12. Home Assignment and Follow Up	Prepare a brief report on the role of comparators in ensuring quality control for aircraft engine components, linking concepts to LP 09 surface finish requirements.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to surface finish, meaning of surface texture and their definition.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min